Photogrammetry of brain dissections is promising, but mesh quality limits quantitative use: lessons from closing 130 BraDiPho meshes

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Impact Statement (40 words max)

Photogrammetric meshes of brain dissections are attractive references for diffusion MRI, but they are rarely closed volumes. We show what automatic closing achieves, where it fails, and why input mesh quality, not closing, limits quantitative use.

Synopsis (100 words max)

Motivation: photogrammetric meshes of fibre dissections could serve as quantitative anatomical references for diffusion MRI, but they are open and defective. Goals: close them automatically into watertight volumes and measure how far these volumes can be trusted. Approach: a repair and capping pipeline (planar caps, curved caps for folded openings) applied to all 130 epochs of the 12 public BraDiPho specimens, checked numerically and visually. Results: all epochs became watertight, and the capping choice changed volumes by 0.45% on average. However, volume increased in 15 of 104 dissection steps, by up to 15%, mostly because the scanned boundary moved.

Introduction

Fibre dissection combined with photogrammetry produces textured 3D meshes of the brain at successive dissection steps ("epochs"). The BraDiPho archive[1] makes them public for comparison with diffusion MRI tractography. Quantitative use (volumes, volume removed per step, registration to MRI) requires watertight surfaces, but the meshes are open on the face that was not photographed and contain scanning defects (duplicated and inverted faces, flaps, non-manifold edges, debris). We closed the entire archive automatically and asked: are the volumes good enough to measure what the dissection removed? We report what works, what does not, and what we wish we had known at the start.

Methods

Data. All 130 epochs of the 12 BraDiPho specimens; "b" variants, whose meaning is undocumented, were excluded from comparisons between epochs.

Pipeline (Python, VTK, Triangle). (1) Repair: duplicated faces, small flaps and open fragments removed; non-manifold edges split; orientation made consistent and outward. (2) Boundary loops extracted. (3) For each loop, a plane is fitted by SVD. If the projected loop is a simple polygon (after straightening micro-folds of up to 20 vertices), a planar cap is triangulated (constrained Delaunay) on the outer side of the opening and joined to the loop by a side band; otherwise, a curved cap is built by harmonic interpolation. (4) Final repairs, removal of debris inside or far from the brain, outward orientation of each part. (5) Volume is computed only if no boundary or non-manifold edge remains.

Evaluation. (a) Watertightness. (b) Sensitivity of volume to capping: comparison with a flat-fan closing, and two fold-straightening variants (A: always planar; B: planar only if the added wedge is $\leq$5%). (c) Visual inspection of every epoch. (d) Plausibility: dissection only removes tissue, so volume should decrease. Each increase was compared with control pairs for scale (ICP with scaling), position of the scanned boundary, and cap wedge.

Results

What works. All 130 epochs became watertight (Fig. 1). Straightening micro-folds reduced epochs needing a curved cap from 38 to 12. Volume differed from the flat-fan closing by 0.45% on average (maximum 3.4%). Visual defects dropped from 29 to 12 epochs.

What works poorly. The 12 remaining defective epochs (7 from one specimen) have truly folded openings: the curved cap folds or spans beyond the edge (Fig. 3a). Small holes just under a planar cap leave inverted patches (Fig. 3b). On four hollowed openings, variants A and B disagreed by 5.7–11%: volume there depends on an arbitrary choice.

What does not work. Over 104 consecutive-epoch pairs, the median volume removed was 8.7% (0.5–37.5%), but volume *increased* in 15 pairs, by up to 15.3%, and 21 changes were within capping uncertainty (Fig. 2). Epochs share one frame (median distance 0.4–0.7 mm); scale differences ($\leq$0.8%) did not distinguish increases from controls. In 9 of 15 increases, including the three largest, the scanned boundary lay 2.2–5.1 mm further out, adding 19–57 cm^3 (Fig. 3c). Two followed poorly closed epochs; four were within uncertainty.

Discussion

"Watertight" does not mean "correct": a three-vertex micro-fold suffices to replace a planar cut by a warped membrane, and inverted debris silently subtracts volume. Only visual inspection revealed these defects. Capping uncertainty (about 3%, up to 11% on hollowed openings) cannot explain increases of 13–15%: the scanned boundary itself moves, so any closing would inherit the shift. Both faces are dissected, so an inward shift can be real, but an outward shift of 2–5 mm cannot: it reflects tissue settling, slight specimen motion and independent captures. Even between control epochs the cap moved by about $\pm$2 mm, and 1 mm costs 2–4% of volume, comparable to a typical dissection step, while tissue removed on the unphotographed face is not measured at all. The limit is the photogrammetric input, not closing.

Recommendations. Photograph both faces at every epoch; fix the specimen on a rigid support with fiducial markers and a scale reference; reprocess the original photographs consistently; consider LiDAR or structured light for future acquisitions; report capping uncertainty and visual checks with every volume; document the "b" epochs.

Limitations. No ground-truth volumes were available, and real tissue removal on the unphotographed face cannot be separated from settling or motion with the current data.

Conclusion

Photogrammetric dissection meshes can be closed automatically, but their volumes are trustworthy only to within a few percent, and many dissection steps are not measurable. Quantitative use requires better control upstream, at acquisition and reconstruction.

Figures

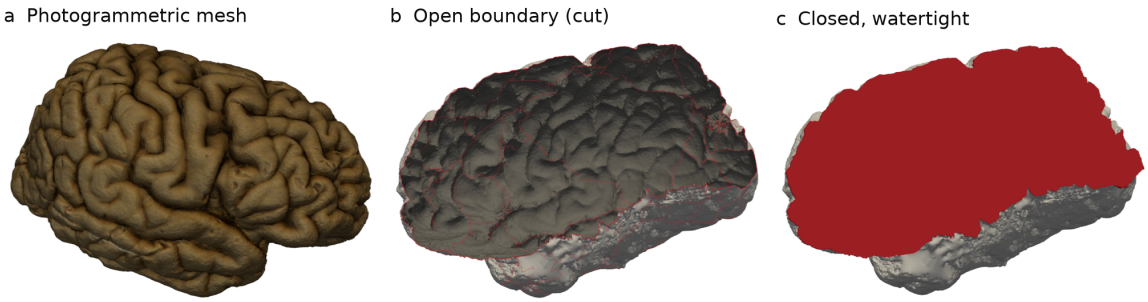


Figure 1: Pipeline on a successful epoch (spc-01, epoch 1). (a) Textured photogrammetric mesh. (b) Open boundary, seen from the unphotographed face (inside of the shell in dark grey). (c) Watertight result, caps in red.

References

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2. Shewchuk JR. Triangle: Engineering a 2D quality mesh generator and Delaunay triangulator. Applied Computational Geometry, LNCS 1148, 1996:203–222.

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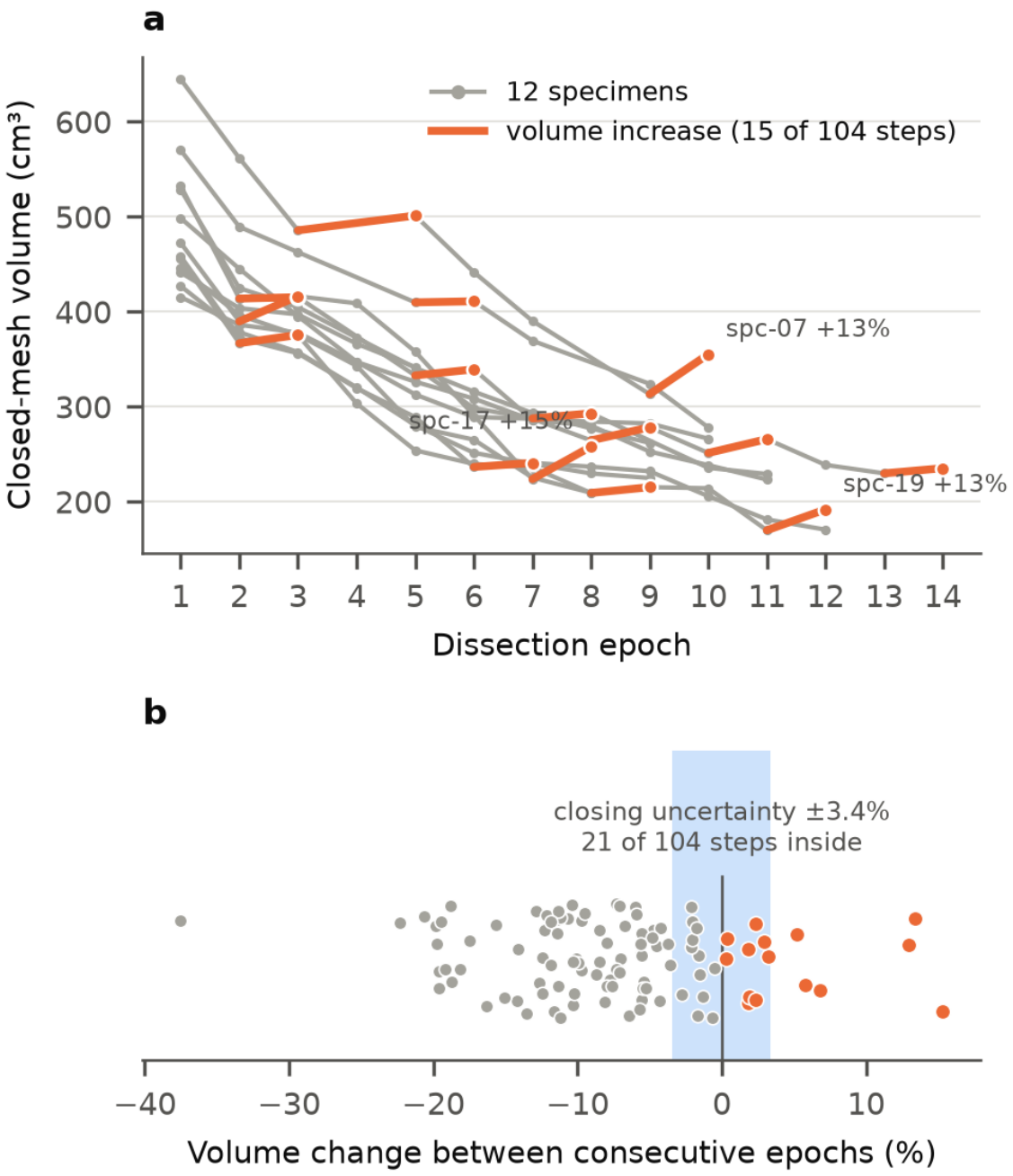


Figure 2: (a) Closed volume versus dissection epoch for the 12 specimens. Volume should only decrease; orange steps increase (15 of 104, up to 15%). (b) Volume change of each step; the blue band is the capping uncertainty (±3.4%), which contains 21 steps.

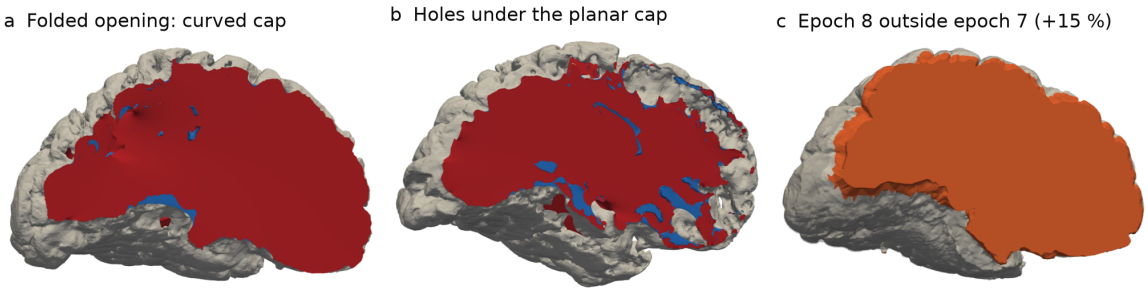


Figure 3: What does not work (caps in red, back-facing surfaces in blue). (a) Folded opening (spc-07, epoch 2): the curved cap folds. (b) Small holes just under the planar cap (spc-15, epoch 8). (c) Surface of epoch 8 lying outside epoch 7 (spc-17, orange): the scanned boundary moved outward, volume +15%.